

Prim's Algorithm

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```
#include <stdio.h>
```

```
#include <time.h>
```

```
int minkey (int key[], int set[], int n)
```

```
{
```

```
    int min = 999, min_idx;
```

```
    for (int i = 0; i < n; i++) {
```

```
        if (set[i] == 0 && key[i] < min) {
```

```
            min = key[i];
```

```
            min_idx = i;
```

```
        }
```

```
    }
```

```
    return min_idx;
```

```
}
```

```
int prim (int graph[][100], int n) {
```

```
    int key[100], set[100], parent[100], mincost = 0;
```

```
    for (int i = 0; i < n; i++) {
```

```
        key[i] = 999;
```

```
        set[i] = 0;
```

```
    }
```

```
    key[0] = 0;
```

```
    parent[0] = -1;
```

```
    for (int k = 0; k < n-1; k++) {
```

```
        int u = minkey (key, set, n);
```

```
        set[u] = 1;
```

```
        for (int i = 0; i < n; i++) {
```

```
            if ((graph[u][i] && set[i] == 0) &&
```

```
                graph[u][i] < key[i]) {
```

```
                parent[i] = u;
```

```
                key[i] = graph[u][i];
```

```
            }
```

```
        }
```

```

for (int i = 1; i < n; i++) {
    printf("%d - %d + %d\n", parent[i], i, graph[parent[i]][i]);
    mincost += graph[parent[i]][i];
}
printf("The minimum cost is %d", mincost);
return 0;
}

```

```

int main() {
    int graph[100][100], n;
    printf("Enter the no. of vertices: ");
    scanf("%d", &n);
    printf("\nEnter the adjacency matrix");
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            scanf("%d", &graph[i][j]);
        }
    }
}

```

clock = start, end;

~~start = clock();~~

prin(graph, n);

end = clock();

double time = (double)(end - start) / CLOCKS_PER_SEC;

printf("Execution time: %f", time);

}

Output:

Enter no. of vertices: 5

Enter the adjacency matrix:

0 2 0 6 0

2 0 3 8 5

0 3 0 0 7

6 8 0 0 9

0 5 7 9 0

